

SEEKRA Project Brief — v19 (Session Handover Edition)

Purpose of this document: complete project state as of 2026-07-31. If a chat session hits its memory limit, paste this brief into a new session to resume work immediately. It captures the live system, the architecture, the deploy workflow, everything built through Step 18B9, every hard-won technical lesson, and the approved roadmap.

1. Product Overview

Seekra — a bilingual (English/Arabic, RTL-aware) Content-Aware Intelligence platform for a media/production organization. It indexes documents, images, audio, and video; provides hybrid search, natural-language visual search, image-to-image similarity, AI chat with citations, PII governance, role-based access, and — since Step 18 — collection-based access control.

Live system: <https://ai.genoa-design.com> (AWS EC2, Nginx HTTPS)

Credentials (development):

- Admin UI/API: admin / admin2026
- Demo viewer: demo_news / demo2026 (user id 4; granted read on “News Production” collection)
- Other users: a-viewer (id 2), a-auditor (id 3)

API login is FORM data, not JSON:

```
TOKEN=$(curl -s -X POST https://ai.genoa-design.com/api/auth/login \
-d 'username=admin&password=admin2026' | python3 -c 'import sys,json;print(json.load(sys.stdin)
↳ ["access_token"]')
```

2. Architecture

- **Backend:** FastAPI (Python 3.11 container), PostgreSQL + pgvector, Celery + Redis (processing pipeline + caches), MinIO object store
- **Frontend:** Next.js 14, static export (npm run build), served via Nginx; i18n via JSON dictionaries (frontend/src/i18n/dictionaries/en.json, ar.json)
- **LLM:** Ollama on the host (chat, summaries); bge-m3 embeddings (1024-dim); visual pipeline: OCR, object labels, visual embeddings, PDF-page and video-keyframe indexing
- **Roles:** admin, editor, viewer, auditor (read-only oversight)
- **GitHub:** rahmirzconsulting-ai/genoa-platform; stable-point backup branches (backup/step18-stable-20260731, backup/pre-step12a-...)

3. Deploy Workflow (proven, use exactly this)

All changes are delivered as **base64 .txt payload packages** containing a tar.gz with an idempotent Python apply script (and any full-file installs):

```
cd ~/genoa/tmp
base64 -d <Package>.txt | tar -xzf -
python3 <apply_script>.py --project /home/ubuntu/genoa
# backend changes:
cd /home/ubuntu/genoa && docker compose up -d --build backend # or full: docker compose up -d --build
```

```
# frontend changes:
cd /home/ubuntu/genoa/frontend && npm run build           # then hard browser refresh
```

Rules:

- Apply scripts must be **idempotent** (done-needle checks → [SKIP]), py3.11-compiled (container is 3.11, sandbox is 3.12), make `.bak_<step>` backups before overwriting, and end with needle verification.
- **Anchor patches preferred** over full-file installs; full-file installs must be rebuilt from verified-live files or their `.bak_*` backups (a stale staging copy once clobbered `warm_visual_model` and crash-looped the backend).
- Verification after every deploy: live curl against the API (payload keys, counts per user role), plus grep of built frontend chunks `/_next/static/chunks/*.js` for `il8n` string literals (component code minifies, strings survive).

4. Step 18 Feature Set (current state, all live-verified)

4.1 Collection permissions (18A/18A2)

- **Model (final): single-collection.** Each file has `collection_id` NULL (root = visible to everyone) or points to one collection (visible only to admins + users with a `can_read` grant in `user_permissions`). Multi-collection tagging was explicitly evaluated and **rejected by the user** (“complex process and no real benefit”) — do not revive the 18C junction-table design.
- **Shared enforcement:** `app/services/permissions.py` — `permission_clause(user, params, document_alias)` returns `1=1` for admin/anonymous, else `(d.collection_id IS NULL OR d.collection_id IN (SELECT collection_id FROM user_permissions WHERE user_id = :perm_uid AND can_read))`. Applied in: files list, classic search `_build_filters`, smart search `_filter_clause` (via `current_search_user` `ContextVar`), and **all chat queries including the visual legs** (OCR/object/vector) and inventory/lib-count queries.
- **Admin UI:** `/admin/permissions/` — grant/revoke; grantable users = non-admin roles.
- `user_permissions` columns: `id`, `user_id`, `collection_id`, `permission_flags`, `expires_at`, `created_at`, `can_read`, `can_write` (`can_read/can_write` were added live via `ALTER TABLE` — see §6 schema-drift lesson).

4.2 Collection management (18B)

- Backend: `PATCH /api/collections/{id}` (rename/department/description), `DELETE /api/collections/{id}?force=` — 409 guard when files inside; `force=true` moves files to root, deletes grants, invalidates `seekra:files:list:*`, audits `SECURITY_COLLECTION_UPDATED/_DELETED`.
- Admin UI `/admin/collections/`: table with file counts, create form, edit modal, delete modal with move-to-root checkbox. EN/AR.

4.3 Files page (18A→18B9)

-  **Collection badge** with collection name on restricted files; `collection_id` merged into the list payload via raw SQL (the Document ORM does **not** map that column) and declared on `DocumentResponse` (FastAPI strips undeclared keys — two separate bugs, 18B4+18B5).
- **Real pipeline status** (18B6): backend sends `status/stage/progress` (raw SQL, same merge query); page shows Processed/Processing/Pending/Failed. Previously status was inferred from

content_text truthiness → text-less images showed “pending” forever despite being completed.

- **Single move-to-collection button** (18B7): permanent in the toolbar next to search; turns primary + shows selection count when files are ticked; dialog works from zero selection with an explanatory hint. Bulk bar keeps delete/clear only.
- **Useful file icon** (18B6): real image thumbnails for image files; clickable → opens viewer for all types.
- Bulk select/move/delete with i18n, PII badge, download, quick preview.

4.4 Upload page (18B7)

- Business-friendly pipeline stage labels EN/ AR: extracting→“Reading your document”, ocr→“Scanning text from images”, transcribing→“Listening and transcribing”, chunking→“Organizing content”, embedding→“Making it searchable”, summarizing→“Writing a summary”, ready→“Ready to search”. Keys `upload.stage.*` in both dictionaries.

4.5 Worker autolink fix (18B8)

- `celery_tasks._link_default_collection` (prototype leftover) silently assigned every processed file to the lowest-id collection (`SELECT id FROM collections ORDER BY id LIMIT 1`) — mis-tagged `logo.png`, `og-image.jpg`, `robots.txt` to “News Production” after collection id 1 was deleted. Now a no-op; new uploads stay at root.

4.6 Governance/audit (18B9)

- **Backend audit coverage verified live complete** for all Step 18 actions: `SECURITY_COLLECTION_CREATED/UPDATED/ DELETED`, `SECURITY_PERMISSION_GRANTED/ REVOKED`, `CONTENT_BULK_MOVE/ DELETE` — all recorded with actor + detail (visible in `/api/audit/`).
- Governance page event lists updated to document all four new event types.

5. Live DB Snapshot (2026-07-31)

- **Collections:** 2 News Production, 3 Documentaries, 4 Commercials, 5 Archive Footage, 7 Legal Documents, 8 Human Resource (“Human Recourse” as typed by user)
- **Files:** ~20; 9+ in collections, rest at root
- **Grants:** `demo_news(4)→News Production(2)` read; demo grants on 3/7 for user testing were added/revoked during verification
- **Caches:** files list cache key version `seekra:files:list:v5`: (bump on every payload-shape change)

6. Hard-Won Technical Lessons (read before any backend change)

1. **Document ORM does not map `collection_id`** (model predates collections). `doc.collection_id` → `AttributeError` → 500. Always raw SQL: `SELECT id, collection_id FROM documents WHERE id = ANY(CAST(:ids AS int[]))` and merge into payload dicts. Defensive `getattr(doc, "pii_flagged", ...)` pattern used for the same reason.
2. **FastAPI `response_model` silently strips undeclared keys.** `list_files` merges `collection_id/status` into dicts, but they vanish unless declared on `DocumentResponse` (`Optional[...] = None`). This caused two “invisible” bugs in a row (18B5, 18B6).
3. **Files-list cache:** key = `seekra:files:list:vN`: + `stable_hash(folder, uid, role)`. **Bump vN on every payload-shape change** and/or flush `seekra:files:list:*` in Redis after deploys, or stale

payloads mask the fix. `_invalidate_files_cache()` exists and is called on upload/move/delete/collection-delete.

4. **MissingGreenlet:** capture `user_id/username` into locals BEFORE commit/rollback; touching expired ORM objects after triggers sync refresh outside greenlet → 500.
5. **Live schema drift:** live `user_permissions` lacked `can_read/can_write` (had `permission_flags`). Fixed in-container: `docker compose exec backend python3 -c using os.environ['DATABASE_URL']` with ALTER TABLE. Treat ORM assumptions as suspect; check real columns first.
6. **Stale staging hazard:** never full-file-install from an old staging copy. Rebuild from live `.bak_*` backups or use anchor patches. (The chat.py incident: 659-line stale file clobbered the 1024-line live file with `warm_visual_model` → ImportError crash-loop.)
7. **f-string braces in generated code:** `f"{indent}}}"` → syntax error; escape as `}}}`.
8. **Container reality check:** `docker compose exec backend grep -n <needle> /app/...` proves whether the running container has new code (host file container image). Docker layer caching can silently serve old code.
9. **Worker leftovers:** the pipeline worker carries prototype-era helpers (`_link_default_collection`); check `celery_tasks.py` for side effects before assuming the API layer is the only writer.
10. **Verification pattern that works:** curl the live API per role (admin vs demo_news), print payload keys/counts; grep built JS chunks for i18n literals; compare `created_at` vs `updated_at` to distinguish insert-time vs update-time effects.

7. Development History (compact)

- **Steps 12–13:** RBAC 4 roles, auditor read-only, governance page, PII masking/reporting, visual OCR/object labels/visual embeddings, natural-language visual search, image-to-image similarity.
- **Step 14–16:** unified smart search, chat visual evidence legs, viewer unification, PDF-page/video-keyframe search, upload live progress.
- **Step 17:** performance (caching, stable_hash keys), bronze Seekra identity polish.
- **Step 18 (this session):** collection permissions end-to-end (18A/18A2 hotfixes A2B/A2C), schema-drift repair, collections management UI+API (18B), collection badge (18B1–18B5: ORM unmapped column + response_model stripping + cache staleness), real status/thumbnails/single move button/friendly stages (18B6/18B7), worker autolink fix (18B8), governance event lists (18B9). Multi-collection design (18C) built then **explicitly rejected** — do not deploy.

8. Known Limitations / Deferred

- Analytics aggregate counts are **not** permission-filtered (admin-only page, v1 — acceptable for now).
- `can_write` exists in schema but is not enforced; only admins upload today.
- Alert-notification delivery: rejected by user. Arabic AI visual labels: pipeline-level, open.
- Optional Files-page collection filter dropdown: offered, not requested.
- Docker images are CPU-optimized; Ollama on host.

9. Roadmap (from brief v18 §19, re-prioritized)

1. **R4 — Production identity hardening (recommended Step 19, launch blocker):** login rate limiting, session policy, security headers, admin password rotation, optional SSO/OIDC.
2. **R2 — Audit retention & export:** retention policy, CSV export, date filters, severity grouping.
3. **R5 — Scale & observability:** query-timing dashboards, cache-hit metrics, slow-query reports, optional GPU eval.

4. **R1 — PDF.js v4 upgrade** + bilingual regression.
5. **R6 — Brand/legal (user side)**: domain registration, GCC trademark, final production URL.

10. Operational Runbook

```
# Live API smoke check
TOKEN=$(curl -s -X POST https://ai.genoa-design.com/api/auth/login -d
↪ 'username=admin&password=admin2026' | python3 -c 'import sys,json;print(json.load(sys.stdin)
↪ ["access_token"])' )
curl -s https://ai.genoa-design.com/api/files/ -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
↪ | head -40

# Audit trail
curl -s "https://ai.genoa-design.com/api/audit/actions" -H "Authorization: Bearer $TOKEN"
curl -s "https://ai.genoa-design.com/api/audit/?limit=200" -H "Authorization: Bearer $TOKEN"

# Backend logs / container code check
cd /home/ubuntu/genoa && docker compose logs --tail=100 backend
docker compose exec backend grep -n "<needle>" /app/app/api/files.py

# Flush files-list cache
docker compose exec redis redis-cli --scan --pattern 'seekra:files:list:*' | xargs -r docker compose
↪ exec -T redis redis-cli DEL

# Full local backup (DB + MinIO + code): ~/seekra_full_backup.sh (creates ~/backups/
↪ seekra_full_<ts>.tar.gz)
# GitHub backup: branch backup/<label> on rahmirzconsulting-ai/genoa-platform; ensure .env,
↪ node_modules, tmp/, *.bak_* are gitignored
```

11. How to Resume in a New Session

1. Paste this brief (or attach SEEKRA_Project_Brief_v19.md) and say: “Resume Seekra from brief v19.”
2. State the next step (recommended: **Step 19 — R4 identity hardening**).
3. The assistant should verify live state with the smoke check above before changing anything, deliver changes as idempotent base64 apply packages per §3, and live-verify after each deploy.

End of SEEKRA Project Brief v19